PRACTICAL : 05

ONE AND PAIRED SAMPLE TESTS : WILCOXON SIGNED - RANK TEST AND KS ONE SAMPLE GOODNESS OF FIT TEST.

QUESTION : 01

The null hypothesis is

$$H_0 : \mu = 20$$
$$H_1 : \mu > 20$$

Let μ be average waiting time by scales representative

To calculate the test statistic we need to construct following table.

x_i	$x_i - \mu_0$	$d_i = \lvert x_i - \mu_0 \rvert$	Rank
25	5	5	2.5
10	-10	10	7.5
15	-5	5	2.5
20	0	0	—
17	-3	3	1
11	-9	9	6
30	10	10	7.5
27	7	7	5
36	16	16	10
40	20	20	11
5	-15	15	9
26	6	6	4

Here sum of positive rank $P^+ = 40$

Sum of negative rank $P^- = 26$

$n = 11$

$$T = \min(T^+ T^-)$$
$$= \min(40, 26) = \underline{\underline{26}}$$

$$T_{(11, 0.05)} = 14$$

Conclusion

Here $T_{cal} >$ Table value

$\Rightarrow 26 > 14$.

Hence we reject H_0

Question : 02

Here hypothesis is given by.

$H_0 : \mu = 119$

$H_1 : \mu \neq 119$

To calculate the test statistic we need to construct following table

x_i	$d_i = \lvert x_i - \mu_0 \rvert$	Rank
121	2	1.5
115	4	3.5
79	40	11
52	67	15
102	17	9
126	7	6
81	38	10
65	54	13
109	10	7
119	0	
115	4	3.5
121	2	1.5
103	16	8
75	44	12

59	60	14
125	6	5

$T = \min(14, 106)$
$= 14.$

$T_{15, \frac{0.05}{2}} = 25$

$T < T_{15, 0.025}$

CONCLUSION.

Here $T < T_{15, 0.025}$

Hence we reject H_0

QUESTION : 03

Here the hypothesis is given by.

$H_0 : \mu = \mu_0$
$H_1 : \mu < \mu_0$

To calculate the test statistic we need to construct following table

x_i	y_i	$d_i = \lvert x_i - y_i \rvert$	Rank
40	45	5	9.5
45	50	5	9.5
60	55	5	9.5
50	52	2	2.5
55	52	3	5.5
60	60	0	—
78	55	23	18
80	53	27	19
65	70	5	9.5
62	65	3	5.5

70	68	2	2
71	50	21	17
65	58	7	14
45	47	2	2.5
48	54	6	13
42	50	8	13.5
48	43	5	9.5
68	60	8	13.5
50	48	2	2.5
55	50	5	9.5

The sum of positive rank = 122.5
The sum of negative rank = 67.5

$T_{19,0.05} = 54$

Conclusion

$T = \max(122.5, 67.5)$

$= \underline{122.5}$

$T^{+} > T_{19,0.05}$

$\Rightarrow 122.5 > 54$

Hence we accept H_0.

QUESTION : 04.

Here the hypothesis is given by

$H_0 : \mu_1 = \mu_2$

$H_1 : \mu_1 \neq \mu_2$

Before	After	$d_i = \lvert X_1 - Y_1 \rvert$	Rank
72	74	2	8
70	72	2	8
68	69	1	3.5
67	68	1	3.5
73	72	1	3.5
71	72	1	3.5
72	72	0	

70	71	1	3.5
69	67	2	8
60	73	13	10
68	69	1	3.5

The sum of positive rank = 43.5

The sum of negative rank = 11.5

CONCLUSION

$$T = \max(40, 11.5)$$
$$= 40$$

Hence accept H_0 since $H_0 : \mu_1 = \mu_2$.

QUESTION : 05

The hypothesis is given

H_0 : data have come from $U(0,1)$

H_1 : not from $U(0,1)$. [$U(0,1)$ does not fit good to data].

if $X \sim U(0,1)$.

$$F_0(x) = \int_0^x f(t)\,dt$$
$$= \int_0^x 1\,dt = [t]_0^x = \underline{\underline{x}}$$

The emphirical distribution is given by.

$$F_n(x) = \begin{cases} 0 & \text{if } X \le X_{(n)} \\ i/n & \text{if } X_{(i)} \le x \le X_{(i+1)} \\ 1 & \text{if } X \ge X_{(n)}. \end{cases}$$

x_i	$F_0(x)=X_{(i)}$	$F_n(x)$	$[F_n(x) - F_0(x)]$
0.44	0.1	0.067	0.03331
0.76	0.13	0.127	0.0033
0.13	0.23	0.2	0.03
0.27	0.27	0.267	0.0033
0.97	0.39	0.333	0.05667
0.45	0.44	0.4	0.04
0.1	0.45	0.467	0.0167
0.94	0.45	0.533	0.0333
0.39	0.5	0.6	0.1
0.53	0.53	0.67	0.1367
0.85	0.76	0.73	0.02
0.45	0.85	0.8	0.05
0.73	0.44	0.867	0.07333
0.98	0.97	0.93	0.03667
0.5	0.98	1	0.02

CONCLUSION

$$D_n = \sup_x [F_n(x) - F_0(x)] = 0.1367$$

$$D_{n,\alpha} = 0.338$$

Hence we Accept H_0

QUESTION 06

H_0: The random variables follows poisson distribution with mean 7.6

Now if $X \sim P(7.6)$ the $F_0(x) = \sum_{z=0}^{x} \frac{e^{-\lambda}\lambda^x}{x!}$

Also emphirical distribution is given by

$$F_n(x) = \begin{cases} 0 & \text{if } z < 0 \\ 1 & \text{if } z \geq \text{maximum}(x) \\ a(x)/N & \text{otherwise} \end{cases}$$

$C_u(x)$ Cumulative frequency of x

x_i	f_i	$C_u(x_i)$	$F_n(x_0)$	$F_n(x_i^+)$	$\lvert F_n(x_i) - F_0(x_i^2) \rvert$
0	0	0	0.000	0.001	0.001
1	5	5	0.001	0.004	0.003
2	14	19	0.006	0.019	0.013
3	24	43	0.013	0.055	0.043
4	37	100	0.030	0.125	0.095
5	111	211	0.063	0.231	0.168
6	197	408	0.121	0.365	0.243
7	278	686	0.204	0.510	0.306
8	378	1064	0.316	0.648	0.332
9	418	1432	0.440	0.765	0.325
10	461	1943	0.577	0.854	0.776
11	433	2376	0.706	0.915	0.209
12	413	2789	0.829	0.954	0.125
13	358	3147	0.935	0.976	0.041
14	219	3366	1.000	0.989	0.011

$$D_n = \sup_x \lvert F_n(x) - F_0(x) \rvert = 0.332.$$

CONCLUSION.

$D_n = 0.332$

$D_{n,0.05} = 0.023 = \frac{1.36}{\sqrt{3366}}$

Hence we reject H_0.

(7) H_0 : Normal distribution is good fit
H_1 : Normal distribution is not good fit

To calculate the test statistic we need to construct following table

i	x	$F_n(x) = i/n$	$Z_i = \frac{x_i - 3}{1}$	$F_0(x) = P(Z < Z_i)$
1	1	1/10	−2	0.0228
2	1.9	2/10	−1.1	0.1352
3	2.1	3/10	−0.9	0.1841
4	2.7	4/10	−0.3	0.3821
5	2.8	5/10	−0.2	0.4207
6	3.2	6/10	0.2	0.5793
7	3.6	7/10	0.6	0.1258
8	3.9	8/10	0.9	0.8154
9	4.2	9/10	1.2	0.8849
10	5.1	10/10	2.1	0.9821

CONCLUSION

$$D_n = \sup |F_n(x) - F_0(x)|$$
$$= 0.1159$$
$$D_{10,\alpha} = 0.410$$

Hence we accept H_0

QUESTION : 08

H_0 : Binomial distribution fits good

H_1 : Binomial distribution is not good fit

To calculate the test statistic we need to construct following table

x_i	f	xf	$f_n(m) = P(X=x)$	$F_n(x)$
0	23	0	0.0616	0.0616
1	95	95	0.2481	0.3097
2	164	328	0.3150	0.6847
3	92	276	0.2519	0.9366
4	26	104	0.0634	1
	400	803		

$$\bar{X} = \frac{\sum x_i}{n} = \frac{803}{400} = 2.0075$$

$$\hat{p} = \frac{\bar{X}}{n} = 0.5019$$

$$\hat{q} = 1 - p = 0.4981$$

$$P(X = x) = {}^{n}C_{x}\, p^{x}\, q^{n-x}$$

For F_0 $\quad p = 0.5 \quad q = 0.5$

$f_0(x) = {}^{4}C_{x}\,(0.5)^{2}(0.5)^{4-x}$	$F_0(x)$	$\lvert F_n(x) - F_0(x) \rvert$
0.0625	0.0625	0.0009
0.25	0.3125	0.0028
0.375	0.6875	0.0028
0.25	0.9375	0.0009
0.0625	1	0

CONCLUSION

$$D_n = \sup \lvert F_n(x) - F_0(x) \rvert = 0.0028$$

$$D_{4,0.05} = 0.486$$

$$D_n < D_{n,0.05}$$

Hence we ~~reject~~ ~~Ho~~. accept H_0